

W06 · M06 · 25 MINIT / MINUTES · Hari 2, 12:00 hingga 12:25

Hands-On SQA-AI Assistant Workshop

AI bug reports and quality metrics

Laporan pepijat dan metrik kualiti dibantu AI

Turn the failure you reproduced into a complete bug report with KRISAv2 severity, compute metrics from your test run, and prepare a triage summary.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run /bug-report with steps, expected and actual result
2. Check severity against KRISAv2 Bab 6 Jadual 6.3
3. Run /metrics with your run numbers
4. Ask for a three-line triage summary

1. Jalankan /bug-report dengan langkah, hasil dijangka dan hasil sebenar
2. Semak severity dengan KRISAv2 Bab 6 Jadual 6.3
3. Jalankan /metrics dengan angka larian anda
4. Minta ringkasan triage tiga baris

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)
- Outputs from earlier workshops exist in outputs\ (or copy them from example-output)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

outputs/m06-bug-report-BUG-CHK-001.md, outputs/m06-metrics.csv

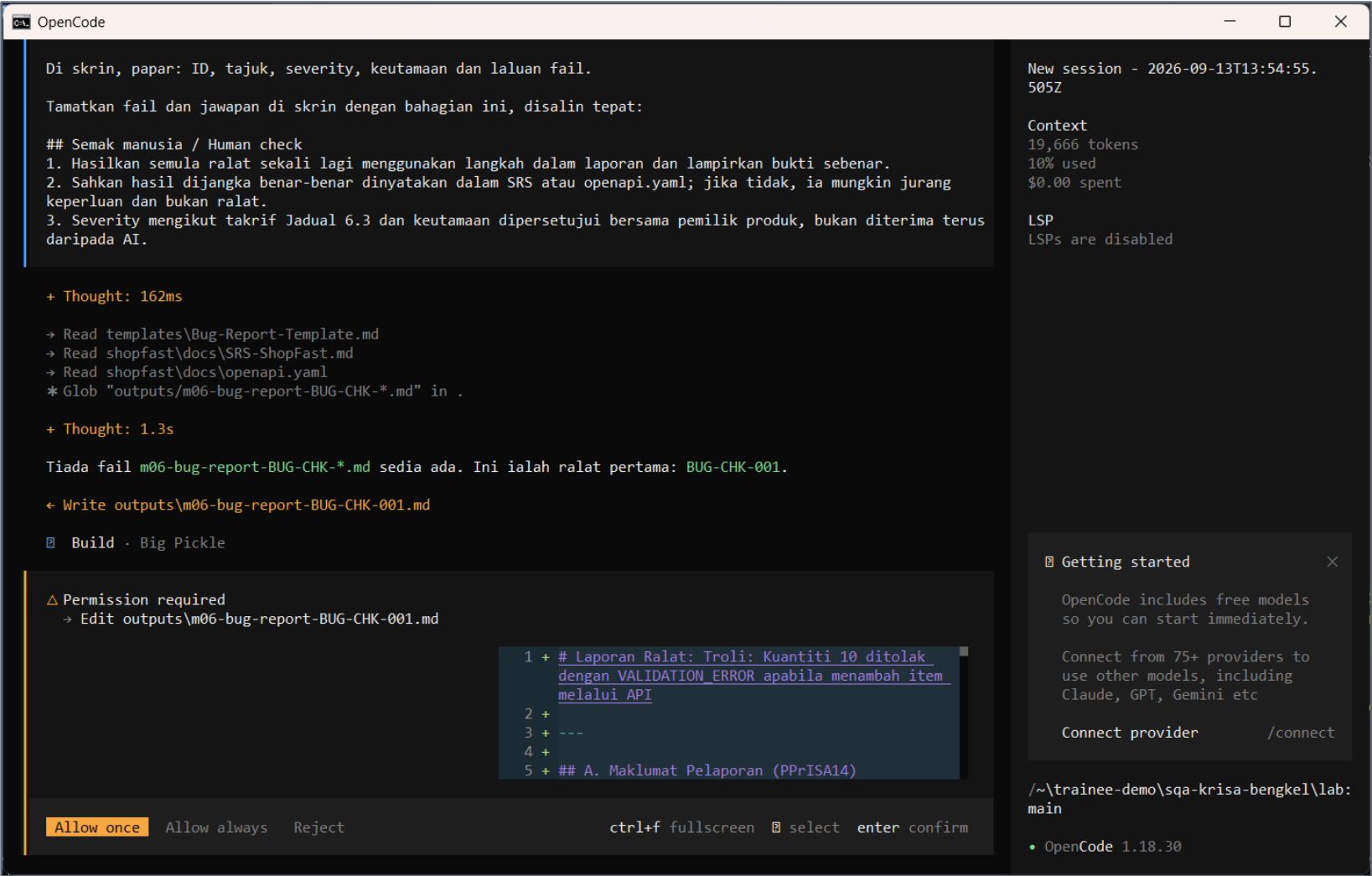
W06 · STEP 1 OF 5 · LANGKAH 1

Run /bug-report with your reproduction evidence. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /bug-report dengan bukti penghasilan semula anda. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

/bug-report Langkah: tambah P002 kuantiti 10 melalui POST /api/cart/items. Dijangka: 201 dan kuantiti 10 diterima (REQ-CHK-01). Sebenar: 400 VALIDATION_ERROR. Bukti: ujian tests/ui/m05-senario.spec.js gagal, binaan 1.0.0.

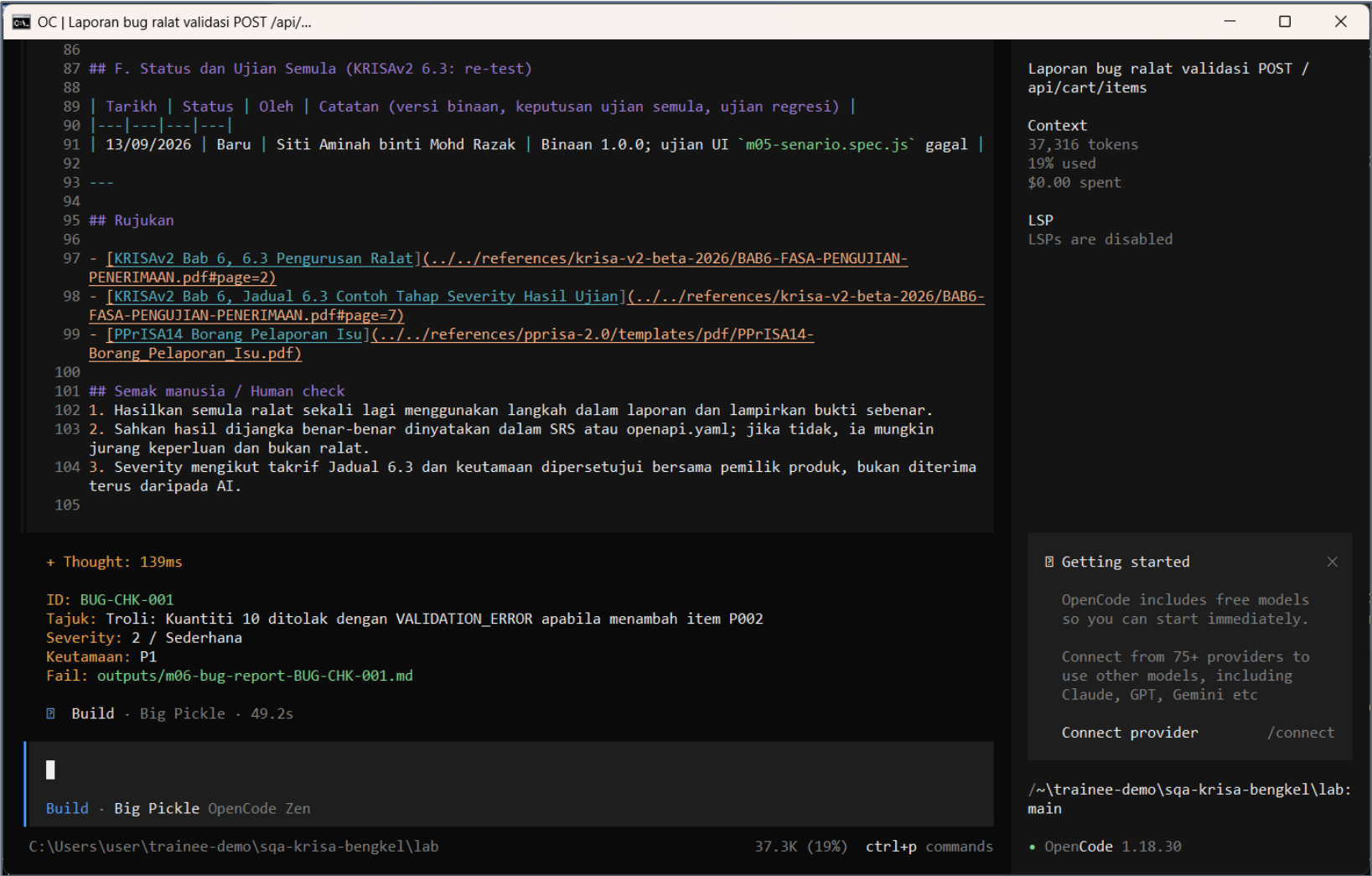


Allow once. Check the severity against KRISAv2 Jadual 6.3.

Allow once. Semak severity dengan KRISAv2 Jadual 6.3.

TYPE IN OPENCODE

Allow once (Enter)



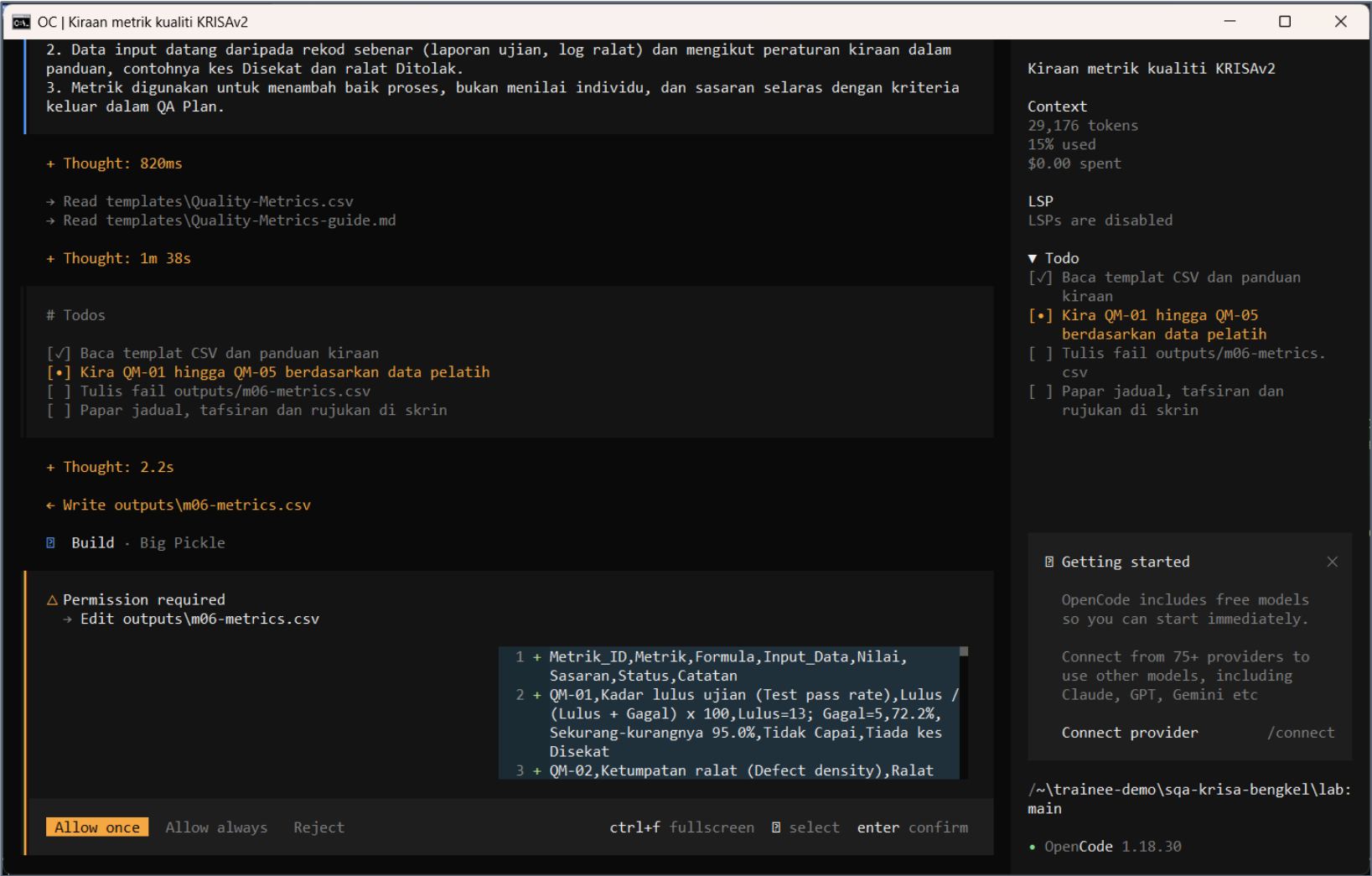
W06 · STEP 3 OF 5 · LANGKAH 3

Run /metrics with your test run numbers. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /metrics dengan angka larian ujian anda. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

/metrics kes dirancang 20, dilaksana 18, lulus 13, gagal 5, pepijat ditemui 5, pepijat bocor ke produksi 1, saiz kod 1.2 KLOC



Allow once. The metrics CSV shows each formula.

Allow once. CSV metrik memaparkan setiap formula.

TYPE IN OPENCODE

Allow once (Enter)

OC | Kiraan metrik kualiti KRISAv2

	ralat (DRE)				
QM-04	Kebocoran ralat (Defect leakage)	Tiada Data	(kosong)	Tidak melebihi 10.0%	Tiada Data
QM-05	Purata masa pembaikan (MTTR)	Tiada Data	(kosong)	Severity 1 tidak melebihi 8 jam	Tiada Data

Input_Data: QM-01: Lulus=13; Gagal=5; QM-02: Ralat=5; Saiz=1.2 KLOC; QM-03: Sebelum=5; Selepas=1. QM-04 dan QM-05 tiada input (tiada pecahan ralat UAT/ujian sistem dan tiada jam pembaikan).

Tafsiran (3 poin)

- QM-01 72.2% jauh di bawah sasaran 95.0%: binaan belum matang dan 5 gagal mesti dikaji punca sebelum gate pelepasan; juga hanya 18 daripada 20 kes dirancang dilaksanakan, jadi liputan tidak lengkap.
- QM-03 83.3% di bawah 90.0%: satu ralat bocor ke produksi. Walaupun nisbah kecil, kualiti serahan belum selaras dengan kriteria keluar, dan ralat produksi itu memerlukan analisis punca akar.
- QM-02, QM-04 dan QM-05 belum boleh dinilai: saiz diberi dalam KLOC sedangkan sasaran dalam UFP, dan tiada data pecahan UAT/ujian sistem serta jam pembaikan. Lengkapkan data ini sebelum keputusan pelepasan.

Rujukan

KRISAv2 Bab 6, 6.6 Metrik Pengukuran (../references/krisa-v2-beta-2026/BAB6-FASA-PENGUJIAN-PENERIMAAN.pdf#page=6) dan PPrISA 2.0, 5.4.2 ii b) Penilaian Kualiti Serahan Projek (../references/pprisa-2.0/PPrISA 2.0_Versi_Beta_Februari_2025.pdf#page=94)

Semak manusia / Human check

- Kira semula sekurang-kurangnya dua metrik secara manual atau dalam Excel dan bandingkan dengan nilai AI.
- Data input datang daripada rekod sebenar (laporan ujian, log ralat) dan mengikut peraturan kiraan dalam panduan, contohnya kes Disekat dan ralat Ditolak.
- Metrik digunakan untuk menambah baik proses, bukan menilai individu, dan sasaran selaras dengan kriteria keluar dalam QA Plan.

Build · Big Pickle · 2m 53s

Build · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab31.0K (16%)ctrl+p commands

Kiraan metrik kualiti KRISAv2

Context31,014 tokens16% used\$0.00 spent

LSP
LSPs are disabled

▼ Todo

- [✓] Baca templat CSV dan panduan kiraan
- [✓] Kira QM-01 hingga QM-05 berdasarkan data pelatih
- [✓] Tulis fail outputs/m06-metrics.csv
- [•] Papar jadual, tafsiran dan rujukan di skrin

Getting started

OpenCode includes free models so you can start immediately.

Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc

Connect provider /connect

/~\trainee-demo\sqa-krisa-bengkel\lab:main

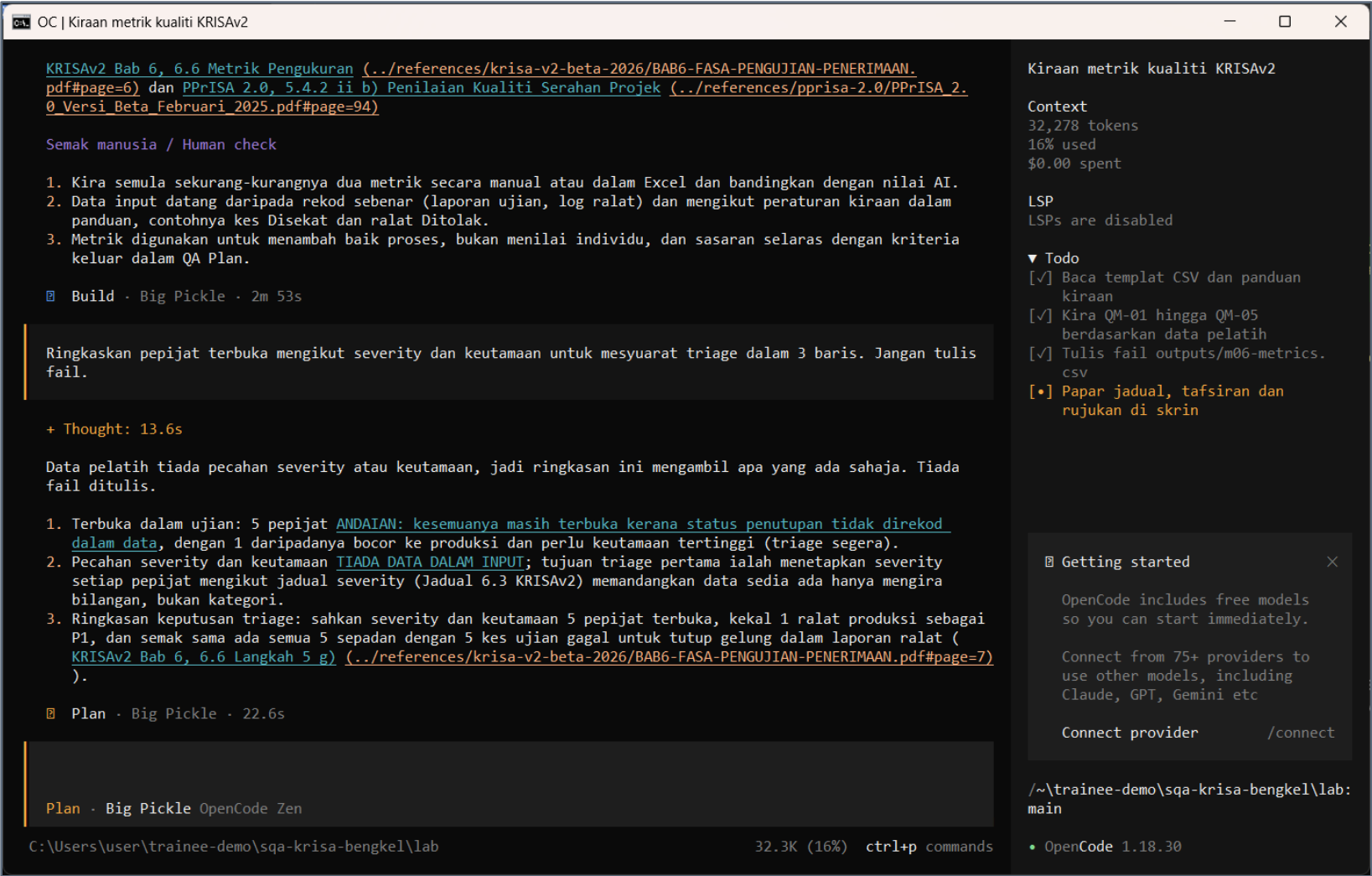
• OpenCode 1.18.30

Switch to Plan and ask for a three-line triage summary.

Tukar ke Plan dan minta ringkasan triage tiga baris.

TYPE IN OPENCODE (PLAN)

Ringkaskan pepijat terbuka mengikut severity dan keutamaan untuk mesyuarat triage dalam 3 baris. Jangan tulis fail.



Human check before you accept the output / Semakan manusia

Steps can be reproduced by someone else

Langkah boleh dihasilkan semula oleh orang lain

Severity follows KRISAv2 Jadual 6.3; priority is a business decision

Severity mengikut KRISAv2 Jadual 6.3; keutamaan ialah keputusan bisnes

Metric formulas are shown so the numbers can be rechecked

Formula metrik dipaparkan supaya angka boleh disemak semula

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting `opencode`

Port 3000 already in use

`$env:PORT = 3001` before `npm start`, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
CSV	Comma-Separated Values	A simple table file that opens in Excel, with commas between the columns. <i>Fail jadual ringkas yang boleh dibuka dalam Excel, dengan koma di antara lajur.</i>
KLOC	Thousand Lines of Code	A size measure used in rates such as defects per KLOC. <i>Seribu baris kod. Ukuran saiz yang digunakan dalam kadar seperti kecacatan bagi setiap KLOC.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>

Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout. <i>Keperluan. Awalan dalam ID keperluan seperti REQ-CHK-01. CHK bermaksud checkout.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>
UAT	User Acceptance Test	Users try the system to confirm it meets their needs before they accept it. <i>Ujian Penerimaan Pengguna. Pengguna mencuba sistem untuk memastikan ia memenuhi keperluan sebelum menerimanya.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 6, 6.3 Pengurusan Ralat \(Rajah 6.2\) \(p.2\)](#)

[Bab 6, Jadual 6.3 Tahap Severity \(p.7\)](#)

[Bab 6, 6.8 Defect Triage dan kriteria keluar UAT \(p.21\)](#)